

BECOMING A DIRECT GOOGLE RCS PROVIDER

The Definitive Tier-1 CPaaS Blueprint & Carrier Integration Playbook

Document Version: 1.0 Enterprise Edition | **Target Spec:** GSMA Universal Profile 2.4 / Google RBM v1
Architected for: Qiyam Ventures & Enterprise Business OS | **Scope:** Zero Third-Party Dependency (Direct Google Cloud & Carrier Interconnect)

Executive Mandate: This engineering blueprint outlines the exact regulatory, architectural, and backend engineering roadmap to operate your own **Direct RCS Business Messaging (RBM) Provider**—the exact same model utilized by global CPaaS operators such as Twilio, Sinch, and Infobip. By integrating directly with **Google Jibe Cloud Hub**, your organization eliminates per-message resale markups, takes direct control over verified sender approvals, acquires wholesale carrier pricing, and can independently issue RCS APIs to enterprise clients.

1. The RCS Ecosystem & How Twilio Actually Operates

A common industry misconception is that companies like Twilio own telecommunication towers or physical base stations to deliver RCS. In reality, RCS is an IP-based data protocol standard defined by the **GSMA (Universal Profile 2.4)**. Google operates the dominant global infrastructure layer known as **Google Jibe Cloud**.

When an Android user opens Google Messages, their device establishes a persistent, TLS-encrypted data socket directly to Google Jibe servers connected with telecom carriers (Reliance Jio, Bharti Airtel, Vodafone Idea in India, and AT&T/T-Mobile/Vodafone globally).

Table 1.1: Direct Google RBM vs. 3rd-Party Resale Aggregators

Metric / Dimension	Using 3rd-Party (Twilio/Sinch)	Direct Google RBM Partner (Your Own)
Cost per Message	Wholesale Telco cost + █0.20–█0.45 markup	Pure Wholesale Telco Rate (0% Markup)
Brand Verification	Depends on aggregator support queue (weeks)	Direct submission to Google Partner console
API Rate Limits	Capped by aggregator account tier	Direct Google Cloud quota (10,000+ QPS)
Data Privacy	Customer chats pass through 3rd-party servers	End-to-End TLS encrypted directly with Google
Commercial Role	Consumer / End-customer	Registered CPaaS Provider & API Reseller

2. Legal, Regulatory & Google Partner Prerequisites

To become recognized as an official RBM Solution Provider by Google, your business entity must complete three specific registrations:

Step 2.1: Corporate D-U-N-S Number (Dun & Bradstreet)

Google requires a valid 9-digit D-U-N-S number to verify legal entity ownership. If your company (Pvt Ltd, LLC, Inc.) does not already possess one, apply through the official Dun & Bradstreet portal (free of cost, takes 3–5 business days). Ensure your corporate legal name, address, and domain match your business filing precisely.

Step 2.2: Google Business Communications Console Enrollment

1. Log into the **Google Business Communications Console** (*business-communications.cloud.google.com*).
2. Select **'Create Partner Account'** and choose the role: **'RCS Business Messaging Solution Provider'**.
3. Provide corporate contact details, registered domain email, business registration documents, and D-U-N-S number.
4. Sign the standard Google RBM Master Services Agreement (MSA). Google's carrier compliance team completes verification within 5–7 business days.

Step 2.3: Telecom Regulatory & DLT Compliance (India & Global)

- **India (TRAI / DLT Regulations):** Register your enterprise on the DLT Portals (Jio DLT, Airtel DLT, Vilpower) as an Enterprise / Telemarketer. Register your Principal Entity (PE ID) and Sender Headers. RCS operates under carrier DLT governance.
- **International (US / EU):** Comply with CTIA opt-out requirements and GDPR user consent mandates. Every promotional rich message must provide a 1-tap compliance opt-out mechanism (which our visual builder automatically includes as the floating Unsubscribe pill).

3. Direct Google Cloud Platform (GCP) Architecture

Once your Google Partner Account is verified, you interact directly with Google Cloud APIs with zero intermediate libraries.

3.1: Enable the Android RBM API & Service Account

1. In your Google Cloud Console, enable the **Android RCS Business Messaging API** (*rcsbusinessmessaging.googleapis.com*).
2. Navigate to **IAM & Admin > Service Accounts** and create a service account named `rbm-gateway-sa`.
3. Assign the role: **RBM Partner Admin**.
4. Generate and download the JSON Private Key (`credentials.json`).

3.2: Native Google OAuth 2.0 JWT Token Generation

Every request to Google's RBM API requires an OAuth 2.0 Bearer Token with the scope `https://www.googleapis.com/auth/rcsbusinessmessaging`. Here is the standard, zero-dependency token generator for your Django/Python backend:

```
# Native Python Google OAuth 2.0 Bearer Token Engine
from google.oauth2 import service_account
from google.auth.transport.requests import Request

SCOPES = ['https://www.googleapis.com/auth/rcsbusinessmessaging']
creds = service_account.Credentials.from_service_account_file(
    '/var/www/whatsq/google_rbm_credentials.json',
    scopes=SCOPES
)
creds.refresh(Request())
google_access_token = creds.token # Valid for 3,600 seconds
```

3.3: Real-Time Inbound & Delivery Tracking via Google Cloud Pub/Sub

Google does not dispatch webhooks to arbitrary public HTTP endpoints. Google publishes all real-time events (inbound customer text, button taps, delivered receipts, read receipts) through **Google Cloud Pub/Sub**:

1. Create a Pub/Sub topic: `projects/YOUR_GCP_PROJECT/topics/rbm-inbound-events`.
2. Add Google's service account (`rbm-pubsub-service-account@google.com`) as a **Pub/Sub Publisher**.
3. Create a **Push Subscription** delivering to: `https://yourdomain.com/api/conversations/rcs/webhook/`.

4. Gateway Engine Implementation: The 4 Core Endpoints

To function as an independent CPaaS like Twilio, your backend gateway must execute four core primitives:

Endpoint 1: Real-Time RCS Capability Check

Before sending rich media, query Google Jibe to verify whether the recipient's phone number and SIM currently support RCS Universal Profile 2.4. If RCS is disabled, your engine immediately falls back to SMS.

```
POST https://rcsbusinessmessaging.googleapis.com/v1/phones/+919447122334/capability
Authorization: Bearer
Content-Type: application/json

{
  "requestId": "cap-check-99214"
}
```

Endpoint 2: Standalone Rich Card & Universal Action Buttons Dispatch

This dispatches the exact rich card layout showcased in your ICICI Bank & Kotak811 reference messages with full-width action buttons.

```

POST https://rcsbusinessmessaging.googleapis.com/v1/phones/+919447122334/agentMessages?agentId=qiyam-r
bm-agent
Authorization: Bearer
Content-Type: application/json

{
  "contentMessage": {
    "richCard": {
      "standaloneCard": {
        "cardOrientation": "VERTICAL",
        "cardContent": {
          "title": "Big plans for your business? Let our Gold Loan help! ■",
          "description": "■Funds for business? Sorted\n■Gold ownership? Untouched\n■Disbursal? Quick"
        },
        "media": {
          "height": "MEDIUM",
          "contentInfo": { "fileUrl": "https://storage.googleapis.com/assets/banner.jpg" }
        },
        "suggestions": [
          {
            "action": {
              "text": "■ Explore now!",
              "postbackData": "ACTION_EXPLORE",
              "openUrlAction": { "url": "https://icicibank.com/gold-loan" }
            }
          },
          {
            "action": {
              "text": "■ Call Branch Manager",
              "postbackData": "ACTION_CALL",
              "dialAction": { "phoneNumber": "+9118001080" }
            }
          }
        ]
      }
    }
  }
}

```

The 7 Supported GSMA Action Button Types (Covered in Full)

1. openUrlAction: Opens website or app deep link (Globe icon ■).
2. dialAction: Launches native dialer with pre-populated phone number (Phone icon ■).
3. clipboardAction: 1-tap copies discount coupon code, OTP, or account number to device clipboard (Copy icon ■).
4. viewLocationAction: Displays exact GPS coordinates or address pin in Google Maps (MapPin icon ■).
5. createCalendarEventAction: Prompts user to save appointment or webinar to Google Calendar (Calendar icon ■).
6. reply: Quick reply chip that returns payload to chatbot without typing.
7. unsubscribe: Mandatory compliance opt-out pill preventing spam penalties.

5. Wholesale Carrier Pricing & CPaaS Profit Margins

By connecting directly to Google RBM as a Tier-1 Solution Provider, you eliminate third-party markup. Google RBM bills based on three standardized conversation tiers:

Billing Tier	Definition & Characteristics	Wholesale Cost (Approx)	Your Resale Price (CPaaS)
Basic Message	Plain text messages up to 160 characters	■0.12 – ■0.15	■0.22 – ■0.28 (~80% margin)
Single Rich Card	High-res banner, formatted copy, action buttons	■0.28 – ■0.35	■0.60 – ■0.75 (~110% margin)
24-Hour Session	Unlimited 2-way messages within 24h of customer reply	■0.45 – ■0.55	■0.95 – ■1.25 (~120% margin)

6. Six-Week Master Implementation Roadmap

Follow this sequential step-by-step roadmap to achieve full production deployment:

Timeline	Core Objectives	Deliverables & Key Milestones
Week 1	Corporate & Legal Setup	Obtain D-U-N-S Number; Register on Telecom DLT Portals (Jio/Airtel); Submit Google Business Communications Partner application.
Week 2	GCP & API Provisioning	Create Google Cloud Project; Enable Android RBM API; Generate Service Account credentials; Configure Cloud Pub/Sub topic and push subscription.
Week 3	Backend Gateway Build	Implement JWT OAuth 2.0 generator; Wire capability check engine; Connect Pub/Sub push receiver in Qiyam OS Django backend.
Week 4	Studio & Card Engine	Validate Standalone Rich Card and Carousel serializers; Verify all 7 action button types; Connect automatic carrier SMS fallback engine.
Week 5	Carrier Verification	Submit Qiyam Agent to Google & Carrier Review; Obtain Verified Checkmark Crest; Perform end-to-end device testing across Jio & Airtel SIMs.
Week 6	Production Go-Live	Switch Qiyam OS Gateway setting to Direct Google RBM Mode; Launch commercial client onboarding and wholesale API key provisioning.

Conclusion & Readiness: Your Qiyam Business OS codebase is already equipped with the Universal Profile 2.4 data types, the interactive Visual Studio, realistic light/dark Android simulator, and the multi-tenant database models. By following this guide to obtain your direct Google Partner credentials, your platform transitions into an independent, tier-1 CPaaS powerhouse.